SIMATS ENGINEERING

ASSESSMENT TOOL 1 — SOLUTIONS & ANALYSIS

COURSE NAME	Data Warehousing and Data Mining	**COURSE CODE**	CSA16
WEIGHTAGE	40%	**TOTAL MARKS**	50 (5 Questions × 10 Marks Equivalent / BL Level 4–5)

COURSE OUTCOME COVERED (CO4): Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Comprehensive Question Interpretations & Answers

Q1. Hospital Patient Segmentation (BL4 – 8 Marks)

Given Dataset & Sum of Squared Errors (SSE):

Patient ID	Age	Monthly Expense (₹)	Health Risk
P1	28	12,000	65
P2	55	25,000	40
P3	42	20,000	75
P4	30	11,000	70
P5	60	28,000	35

K (Clusters)	SSE	Δ SSE (Drop)
1	600	-
2	350	250 (41.7%)
3	**220**	**130 (37.1%) — Elbow**
4	200	20 (9.1%)
5	190	10 (5.0%)

DETAILED ANSWER & ANALYTICAL INTERPRETATION

1. Elbow Point Interpretation & Optimal K Determination:

- **SSE Decrease Analysis:** Moving from $K=1$ to $K=2$ reduces SSE by 250 units. Moving from $K=2$ to $K=3$ yields another major reduction of 130 units. Beyond $K=3$, increasing to $K=4$ drops SSE by only 20 units, and $K=5$ drops it by just 10 units.
- **Optimal Clusters ($K = 3$):** The "elbow point" clearly occurs at $K = 3$, where the rate of SSE decrease abruptly flattens. Adding clusters beyond 3 gives diminishing returns in variance reduction.

2. Impact on Patient Segmentation:

- **Cluster Alignment:** The data naturally splits into 3 distinct operational cohorts:
  - *Low Expense, High Risk (Younger):* P1 & P4 (Age ~29, Expense ~₹11.5k, Risk ~67.5).
  - *Moderate Expense, High Risk (Mid-age):* P3 (Age 42, Expense ₹20k, Risk 75).
  - *High Expense, Low-Moderate Risk (Older):* P2 & P5 (Age ~57.5, Expense ~₹26.5k, Risk ~37.5).

3. Healthcare Decision-Making Implications:

- **Resource Allocation & Targeted Interventions:** Allows preventative care programs targeted at high-risk, low-spending young groups (P1, P4) to prevent future costly emergency hospitalizations.

- **Financial & Insurance Planning:** Tailors health insurance premiums and chronic-disease coverage packages specifically for senior, high-expense groups (P2, P5).

Q2. E-Commerce Order Analysis (BL4 – 8 Marks)

Given Dataset:

Order ID	Product Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

DETAILED ANSWER & HIERARCHICAL CLUSTERING ANALYSIS

1. Cluster Formation Process (Agglomerative Hierarchical Clustering):

- **Distance Matrix Calculation:** Computing Euclidean distance based on numerical Order Value (with category attributes):
  - Distance(O1, O3) = |55,000 - 57,000| = ₹2,000 (Same category: Electronics)
  - Distance(O2, O5) = |25,000 - 27,000| = ₹2,000 (Same category: Clothing)
  - Distance(O4, {O2, O5}) = |30,000 - 26,000| = ₹4,000
- **Dendrogram Step-by-Step Merge:**
  1. **Step 1:** Merge O1 and O3 into Cluster A (Electronics, ~₹56,000).
  2. **Step 2:** Merge O2 and O5 into Cluster B (Clothing, ~₹26,000).
  3. **Step 3:** O4 (Furniture, ₹30,000) joins Cluster B at a slightly higher linkage distance, or forms a standalone mid-range tier.

2. Identification of Appropriate Number of Clusters:

- Cutting the dendrogram at a distance threshold around ₹15,000 yields **2 Primary Super-Clusters** (High-Value Electronics vs. Medium-Value Lifestyle/Apparel) or **3 Specific Clusters**:
  - **Cluster 1 (High-Value Tech):** {O1, O3} — Electronics (~₹56,000)
  - **Cluster 2 (Mid-Value Home):** {O4} — Furniture (₹30,000)
  - **Cluster 3 (Regular Apparel):** {O2, O5} — Clothing (~₹26,000)

3. Significance in Understanding Customer Purchasing Behavior:

- **Price Sensitivity & Purchasing Power:** Electronics buyers exhibit high transaction values (₹55k+), indicating premium buyers requiring high-touch support or financing options (EMI).
- **Cross-Selling & Recommendations:** Clothing and furniture buyers share closer spend magnitude (₹25k–₹30k), enabling bundled lifestyle recommendations.

Q3. Mobile App User Behavior — Fuzzy/Soft Clustering (BL5 – 8 Marks)

Given Dataset:

User ID	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2,000
U2	60	5	5,000
U3	140	22	1,500
U4	50	4	6,000
U5	130	20	1,800

DETAILED ANSWER & SOFT CLUSTERING INTERPRETATION

1. Probability-Based Membership Assignment (Fuzzy C-Means / HAC):

- In soft clustering, each user U_i has a membership degree $w_{ik} \in [0, 1]$ for each cluster k, where $\sum_k w_{ik} = 1$.
- **Cluster Profiles Identified:**
  - *Cluster A ("Power Gamers / Highly Engaged"):* High usage (>120 min), high logins (>18), lower spend (~₹1.8k). Members: U1, U3, U5.
  - *Cluster B ("Whales / High Monetization"):* Low usage (~55 min), low logins (~4.5), high spend (≥₹5k). Members: U2, U4.
- **Membership Vector Example:** User U1 might have membership vector $[P(A)=0.92, P(B)=0.08]$, whereas a hybrid user would exhibit $[P(A)=0.55, P(B)=0.45]$.

2. Advantages of Soft Clustering over Hard Clustering:

- **Captures Overlapping User Behaviors:** Real-world users rarely fit binary categories. Soft clustering handles transitionary users who engage moderately and spend moderately.
- **Prevents Information Loss:** Hard clustering forces U2 (50 min usage, ₹5,000 spend) strictly into one bin, ignoring their secondary traits.
- **Nuanced Marketing Personalization:** Enables dynamic targeting (e.g., sending in-app purchase discount prompts to users who show high engagement probability but moderate spend probability).

Q4. Agricultural Crop Monitoring & Data Normalization (BL4 – 8 Marks)

Given Dataset:

Farm	Rainfall (mm)	Soil Fertility Score (1-10)	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

DETAILED ANSWER & INFLUENCE OF NORMALIZATION

1. Differences Observed Before vs. After Normalization:

- **Without Normalization (Raw Scale):** Distance metrics (e.g., Euclidean distance $d = \sqrt{\Delta Rainfall^2 + \Delta Soil^2 + \Delta Yield^2}$) are completely dominated by **Rainfall** because its numerical range (280–520 mm, range = 240) is orders of magnitude larger than Soil Fertility (3–9, range = 6) and Yield (65–90, range = 25).
- *Unnormalized Clustering Result:* Clusters form almost exclusively based on Rainfall values (Grouping {F1, F4} together and {F2, F5} together), ignoring soil quality and yield.
- **With Min-Max Normalization ($x' = \frac{x - x_{min}}{x_{max} - x_{min}}$):** All attributes are scaled strictly to $[0, 1]$, giving equal weight to all 3 variables.

2. How Normalization Influences Cluster Formation & Reliability:

- **Balanced Feature Weighting:** Prevents high-scale attributes from distorting distance metrics.
- **True Multidimensional Insights:** Post-normalization clustering reveals the true biological relationship: *High Soil Fertility (8-9) & Low Rainfall (280-300mm) yields higher output (85-90 tons)* than High Rainfall with Poor Soil (F2, F5).
- **Reliability Improvement:** Ensures decisions regarding irrigation and fertilization are based on holistic productivity factors rather than single dominant metrics.

Q5. Employee Performance Evaluation & Silhouette Analysis (BL5 – 8 Marks)

Given Dataset & Silhouette Scores:

Employee	Productivity	Attendance (%)	Skill Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Model (K)	Silhouette Score	Evaluation
K = 2	0.48	Moderate Cohesion
K = 3	**0.68**	**Optimal Separation**
K = 4	0.52	Sub-optimal Splitting

DETAILED ANSWER & MODEL SELECTION JUSTIFICATION

1. Silhouette Score Analysis & Selection ($s = \frac{b - a}{\max(a, b)}$):

- The Silhouette Score measures intra-cluster cohesion (a) versus inter-cluster separation (b), ranging from -1 to +1.
- **Comparison:** $K=3$ achieves the highest score of **0.68**, significantly outperforming $K=2$ (0.48) and $K=4$ (0.52).
- **Conclusion: $K = 3$ is the most suitable model**, as it maximizes separation between employee tiers while maintaining high internal consistency.

2. Cluster Grouping Interpretation:

- **Tier 1 — High Performers:** {E1, E5} (Productivity ~90.5, Attendance ~94%, Skill ~87).
- **Tier 2 — Moderate / Core Performers:** {E3} (Productivity 78, Attendance 88%, Skill 72).

- **Tier 3 — Underperformers / Need Support:** {E2, E4} (Productivity ~58.5, Attendance ~70%, Skill ~63).

3. Contribution to Effective HR Performance Evaluation:

- **Targeted Upskilling & Training:** Identifies Tier 3 employees for targeted mentorship programs rather than generic performance warnings.
- **Appraisal & Compensation Structuring:** Provides objective, data-driven justification for promotions and merit bonuses for Tier 1 performers.